Reg. No. | | | | | | | | | | | [22CSBS112]

FIRST SEMESTER B.Tech EXAMINATION, FEBRUARY - 2023

INTRODUCTORY TOPICS IN STATISTICS, PROBABILITY AND CALCULUS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer all questions

PART – A **5 X 2=10**

1.	**a.**	Evaluate $\lim_{x\to 2} \frac{x^2-3x+2}{x^2-x-2}$	L-3	CO1	**02**
	b.	Evaluate the integral $\int (e^{5\log x} + e^{3\log x})dx$	L-3	CO2	**02**
	c.	In a moderately asymmetrical distribution, the mode and mean are 32.1 and 35.4 respectively. Find value of median.	L-2	CO3	**02**
	d.	What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed?	L-3	CO4	**02**
	e.	If X is uniform between -2 and 6 then find mean of X.	L-3	CO6	**02**
		PART - B			**6 X 5=30**
2.	**a.**	Prove that $f(x) = \begin{cases} x^2+1 \text{ when } x<2 \\ 5 \text{ when } x=2 \\ 4x-3 \text{ when } x>2 \end{cases}$ is continuous at $x=2$.	L-3	CO1	**05**
		OR			
	b.	Evaluate $\lim_{x\to 3} \frac{x^2-9}{\sqrt{3x+7}-\sqrt{5x+1}}$	L-3	CO1	**05**
3.	**a.**	Evaluate the integral $\int_0^{\pi/2} \frac{\sin x}{1+\cos x} dx$	L-3	CO2	**05**
		OR			
	b.	Evaluate $\int\int xy\, dx\, dy$ over the positive quadrant of the circle $x^2+y^2=a^2$.	L-3	CO2	**05**
4.	**a.**	Explain the different methods of graphical presentation of statistical data.	L-2	CO3	**05**
		OR			
	b.	Represent the following data by drawing a Histogram:	L-2	CO3	**05**

Variable	100-110	110-120	120-130	130-140	140-150	150-160	160-170
Frequency	11	28	36	49	33	20	8

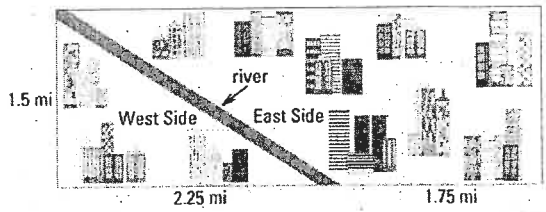
5.	**a.**	There are 10 students of which three are graduates, if committee of five is to be formed, what is the probability that there are (i) only 2 graduates (ii) at least 2 graduates.	L-3	CO5	**05**
		OR			
	b.	A random variable X has the following probability distribution	L-3	CO4	**05**

x	0	1	2	3	4	5	6	7
p(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Find

i. K

ii. P(X<6)

6.	**a.**	Ten coins are tossed simultaneously. Find the probability of getting (i) at least seven heads (ii) exactly seven heads (iii) at most seven heads.	L-3	CO5	**05**
		OR			
	b.	Suppose on an average 1 house in 1000 in a certain district has a fire during a year. If there are 2000 houses in that district, what is the probability that exactly 5 houses will have fire during the year?	L-3	CO5	**05**
7.	**a.**	Define Statistics and write a short note on application of statistics.	L-1	CO3	**05**
		OR			
	b.	You work for a temporary employment agency. You live on the west side of town and prefer to work there. The work assignments are spread evenly throughout the rectangular region shown. Find the probability that an assignment chosen at random for you is on the west side of town. 	L-3	CO6	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	**i.** Differentiate $log\left[\frac{1-x^2}{1+x^2}\right]$. **ii.** If $xy+6y=2x$. *then show that* $\frac{d^2y}{dx^2}=\frac{-24}{(x+6)^3}$.	L-3	CO1	**10**
		OR			
	b.	Find extreme values of $x^3-9x^2+15x-1$.	L-3	CO1	**10**
9.	**a.**	Evaluate the integral $\int\frac{2x^2+5}{2x^2+5x+3}dx$	L-3	CO2	**10**
		OR			
	b.	**i.** Evaluate the integral $\int x^2\sin x\,dx$. **ii.** Find area bounded by the parabola $y^2=16x$ and its latus rectum.	L-3	CO2	**10**
10.	**a.**	**i.** Calculate arithmetic mean by using the short-cut method from the following data:	L-2	CO3	**10**

Marks	20	30	40	50	60	70
No. of Students	8	12	20	10	6	4

ii. The mean marks of 100 students were found to be 40. Later on it was discovered that a score of 53 was misread as 83. Find corrected mean corresponding to the correct score.

		OR			
	b.	Calculate lower and upper quartiles, third decile and 20th percentile from	L-2	CO2	**10**

the following data:

Central value	2.5	7.5	12.5	17.5	22.5
Frequency	7	18	25	30	20

11. a. **i.** One bag contains 4 white balls and 3 black balls, and a second bag contains 3 white balls and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black? — L-3, CO5, **10**

ii. In a class, 70% are boys and 30% are girls. 5% of boys, 3% of the girls are irregular to the classes. What is the probability of a student selected at random is irregular to the classes and what is the probability that the irregular student is a girl?

OR

b. Let X be a discrete random variable with the following probability distribution — L-3, CO4, **10**

X	0	1	2	3
f(x)	0.15	0.38	0.10	0.01

Find Expected value, Variance and standard deviation.

12. a. Weight of ball bearings proved in a certain factory are normally distributed with mean 0.5 Newtons and S.D 0.002 Newtons. Find the percentage of ball bearings produced with weights. — L-3, CO6, **10**

i. less than 0.498 Newtons

ii. more than 0.501 Newtons

iii. between 0.4995 and 0.5005 Newtons

OR

b. **i)** The total duration of football games in the major league in the 2011 (known as the n power Football League) season is uniformly distributed between 447 hours and 521 hours inclusive. — L-3, CO6, **10**

- Find 'a' and 'b' and describe what they represent.
- Find the mean and the standard deviation.
- What is the probability that the duration of games for a team for the 2011 season is between 480 and 500 hours?

ii) The kilometer run (in thousands of k. m) without any sort of problem in respect of a certain vehicle is a random variable having probability density function $f(x) = \begin{cases} \frac{1}{40} e^{-x/40}, & x \geq 0 \\ 0, & x \leq 0 \end{cases}$

find the probability that the vehicle is trouble free

- atmost for 25000 k. m
- Between 16000 to 32000 k. m

13.	**a.**	**i.** If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \cdots \ldots \ldots \infty}}}$. Prove that $\frac{dy}{dx} = \frac{\cos x}{2y-1}$. **ii.** If $x^y = e^{x-y}$. Prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.	L-3	CO6	**10**
		OR			
	b.	From the below given data, does it support the hypothesis that skilled fathers have intelligent girls?	L-3	CO6	**10**

	Intelligent girls	Un-Intelligent girls	Total
Skilled fathers	40	40	80
Unskilled fathers	35	85	120
Total	75	125	200